

Cloud Computing

Lab-5

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SRN:PES2UG23CS906

Class: 6F

Date:12/02/2026

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mona@MonasLaptop:~/PES2UG23CS906$ minikube start
🌈 minikube v1.38.0 on Ubuntu 22.04 (kvm/amd64)
🔧 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📦 Pulling base image v0.0.49 ...
🔄 Updating the running docker "minikube" container ...
🌐 Preparing Kubernetes v1.35.0 on Docker 29.2.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: storage-provisioner, default-storageclass
🎉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
mona@MonasLaptop:~/PES2UG23CS906$ kubectl get nodes
NAME        STATUS   ROLES    AGE   VERSION
minikube    Ready    control-plane  99m   v1.35.0
mona@MonasLaptop:~/PES2UG23CS906$ kubectl create namespace pes2ug23cs906-alpha
namespace/pes2ug23cs906-alpha created
mona@MonasLaptop:~/PES2UG23CS906$ kubectl create namespace pes2ug23cs906-beta
namespace/pes2ug23cs906-beta created
mona@MonasLaptop:~/PES2UG23CS906$ kubectl get namespace
NAME                STATUS   AGE
default             Active   100m
kube-node-lease     Active   100m
kube-public         Active   100m
kube-system         Active   100m
pes2ug23cs906-alpha Active   14s
pes2ug23cs906-beta  Active   10s
mona@MonasLaptop:~/PES2UG23CS906$
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mona@MonasLaptop:~/PES2UG23CS906$ kubectl get namespace
NAME                STATUS   AGE
default             Active   100m
kube-node-lease     Active   100m
kube-public         Active   100m
kube-system         Active   100m
pes2ug23cs906-alpha Active   14s
pes2ug23cs906-beta  Active   10s
mona@MonasLaptop:~/PES2UG23CS906$ kubectl config set-context --current --namespace=pes2ug23cs906-alpha
Context "minikube" modified.
mona@MonasLaptop:~/PES2UG23CS906$ kubectl run citizen --image=nginx --restart=Never
pod/citizen created
mona@MonasLaptop:~/PES2UG23CS906$ kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
citizen     0/1     ContainerCreating   0          5s
mona@MonasLaptop:~/PES2UG23CS906$ kubectl describe pod citizen
Name:         citizen
Namespace:    pes2ug23cs906-alpha
Priority:      0
Service Account: default
Node:         minikube/192.168.49.2
Start Time:   Thu, 12 Feb 2026 14:16:13 +0530
Labels:       run=citizen
Annotations:  <none>
Status:       Running
IP:           10.244.0.6
IPs:          IP: 10.244.0.6
Containers:
  citizen:
    Container ID:  docker://c376f7bf37cfebbfc879366b01bd479d63f49869be88bac1cd5a23560b20c317
    Image:         nginx
    Image ID:      docker-pullable://nginx@sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208
    Port:          <none>
    Host Port:     <none>
    State:         Running
      Started:     Thu, 12 Feb 2026 14:16:19 +0530
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PodScheduled      True
Volumes:
  kube-api-access-shr2v:
    Type:              Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName:      kube-root-ca.crt
    Optional:           false
    DownwardAPI:        true
QoS Class:         BestEffort
Node-Selectors:    <none>
Tolerations:       node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                   node.kubernetes.io/unreachable:NoExecute op=Exists for 300s

Events:
  Type     Reason      Age    From      Message
  ----     -
Normal    Scheduled   10s    default-scheduler   Successfully assigned pes2ug23cs906-alpha/citizen to minikube
Normal    Pulling     9s     kubelet    spec.containers{citizen}: Pulling image "nginx"
Normal    Pulled      4s     kubelet    spec.containers{citizen}: Successfully pulled image "nginx" in 4.916s (4.916s including
waiting). Image size: 160850673 bytes.
Normal    Created     4s     kubelet    spec.containers{citizen}: Container created
Normal    Started     3s     kubelet    spec.containers{citizen}: Container started
mona@MonasLaptop:~/PES2UG23CS906$ kubectl delete pod citizen
pod "citizen" deleted from pes2ug23cs906-alpha namespace
mona@MonasLaptop:~/PES2UG23CS906$

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C:\Users\Monisha>wsl
mona@MonasLaptop:~/PES2UG23CS906$ cd ~/PES2UG23CS906
mona@MonasLaptop:~/PES2UG23CS906$ kubectl run temp --image=nginx --restart=Never
pod/temp created
mona@MonasLaptop:~/PES2UG23CS906$ kubectl delete pod temp
pod "temp" deleted from pes2ug23cs906-alpha namespace
mona@MonasLaptop:~/PES2UG23CS906$

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mona@MonasLaptop:~/PES2UG23CS906$ kubectl get pods -w

^Cmona@MonasLaptop:~/PES2UG23CS906$ kubectl get pods -w
NAME      READY   STATUS      RESTARTS   AGE
temp      0/1     Pending     0           0s
temp      0/1     Pending     0           0s
temp      0/1     ContainerCreating    0           0s
temp      0/1     Terminating        0           6s
temp      1/1     Terminating        0          11s
temp      0/1     Completed            0          12s
temp      0/1     Completed            0          12s
temp      0/1     Completed            0          12s

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pod "temp" deleted from pes2ug23cs906-alpha namespace
mona@MonasLaptopy:~/PES2UG23CS906$ kubectl run citizen --image=nginx --restart=Never
pod/citizen created
mona@MonasLaptopy:~/PES2UG23CS906$ kubectl logs citizen
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/02/12 08:50:33 [notice] 1#1: using the "epoll" event method
2026/02/12 08:50:33 [notice] 1#1: nginx/1.29.5
2026/02/12 08:50:33 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/02/12 08:50:33 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/02/12 08:50:33 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/02/12 08:50:33 [notice] 1#1: start worker processes
2026/02/12 08:50:33 [notice] 1#1: start worker process 29
2026/02/12 08:50:33 [notice] 1#1: start worker process 30
2026/02/12 08:50:33 [notice] 1#1: start worker process 31
2026/02/12 08:50:33 [notice] 1#1: start worker process 32
2026/02/12 08:50:33 [notice] 1#1: start worker process 33
2026/02/12 08:50:33 [notice] 1#1: start worker process 34
2026/02/12 08:50:33 [notice] 1#1: start worker process 35
2026/02/12 08:50:33 [notice] 1#1: start worker process 36
mona@MonasLaptopy:~/PES2UG23CS906$ kubectl run broken --image=busybox --restart=Never --command -- sh -c "echo 'CRASH: something went
wrong' >&2; exit 1"
pod/broken created
mona@MonasLaptopy:~/PES2UG23CS906$ kubectl logs broken
Error from server (BadRequest): container "broken" in pod "broken" is waiting to start: ContainerCreating
mona@MonasLaptopy:~/PES2UG23CS906$ kubectl logs broken
Error from server (BadRequest): container "broken" in pod "broken" is waiting to start: ContainerCreating
mona@MonasLaptopy:~/PES2UG23CS906$ kubectl logs broken
CRASH: something went wrong
mona@MonasLaptopy:~/PES2UG23CS906$

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CRASH: something went wrong
mona@MonasLaptopy:~/PES2UG23CS906$ kubectl create deployment pes2ug23cs906 --image=nginx -n pes2ug23cs906-alpha
deployment.apps/pes2ug23cs906 created
mona@MonasLaptopy:~/PES2UG23CS906$ kubectl get deployments -n pes2ug23cs906-alpha
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
pes2ug23cs906 1/1     1            1           23s
mona@MonasLaptopy:~/PES2UG23CS906$ kubectl get pods -n pes2ug23cs906-alpha
NAME          READY   STATUS    RESTARTS   AGE
broken         0/1     Error     0           2m56s
citizen        1/1     Running   0           3m14s
pes2ug23cs906-7f6477487-rhvbq 1/1     Running   0           36s
mona@MonasLaptopy:~/PES2UG23CS906$ kubectl scale deployment pes2ug23cs906 --replicas=5
deployment.apps/pes2ug23cs906 scaled
mona@MonasLaptopy:~/PES2UG23CS906$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
broken         0/1     Error     0           3m47s
citizen        1/1     Running   0           4m5s
pes2ug23cs906-7f6477487-9xztj 0/1     ContainerCreating 0           5s
pes2ug23cs906-7f6477487-bqc8l 0/1     ContainerCreating 0           5s
pes2ug23cs906-7f6477487-g4cnh 0/1     ContainerCreating 0           5s
pes2ug23cs906-7f6477487-rhvbq 1/1     Running       0           87s
pes2ug23cs906-7f6477487-vhb5x 0/1     ContainerCreating 0           5s
mona@MonasLaptopy:~/PES2UG23CS906$ kubectl scale deployment pes2ug23cs906 --replicas=3
deployment.apps/pes2ug23cs906 scaled
mona@MonasLaptopy:~/PES2UG23CS906$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
broken         0/1     Error     0           4m1s
citizen        1/1     Running   0           4m19s
pes2ug23cs906-7f6477487-bqc8l 0/1     Terminating 0           19s
pes2ug23cs906-7f6477487-g4cnh 0/1     ErrImagePull 0           19s
pes2ug23cs906-7f6477487-rhvbq 1/1     Running       0           101s
pes2ug23cs906-7f6477487-vhb5x 0/1     ContainerCreating 0           19s
mona@MonasLaptopy:~/PES2UG23CS906$ kubectl get pods

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pes2ug23cs906-7f6477487-vhb5x 0/1 ContainerCreating 0 33s
mona@MonasLaptop:~/PES2UG23CS906$ kubectl get rs,po
NAME                                DESIRED    CURRENT    READY    AGE
replicaset.apps/pes2ug23cs906-7f6477487 3          3          3        2m26s

NAME                                READY    STATUS    RESTARTS    AGE
pod/broken                          0/1     Error     0            4m46s
pod/citizen                         1/1     Running   0            5m4s
pod/pes2ug23cs906-7f6477487-g4cnh  1/1     Running   0            64s
pod/pes2ug23cs906-7f6477487-rhvbq  1/1     Running   0            2m26s
pod/pes2ug23cs906-7f6477487-vhb5x  1/1     Running   0            64s
mona@MonasLaptop:~/PES2UG23CS906$ kubectl set image deployment/pes2ug23cs906 nginx=nginx:1.25
deployment.apps/pes2ug23cs906 image updated
mona@MonasLaptop:~/PES2UG23CS906$ kubectl get rs,po
NAME                                DESIRED    CURRENT    READY    AGE
replicaset.apps/pes2ug23cs906-58b8bf4595 1          1          0         8s
replicaset.apps/pes2ug23cs906-7f6477487 3          3          3        2m46s

NAME                                READY    STATUS    RESTARTS    AGE
pod/broken                          0/1     Error     0            5m6s
pod/citizen                         1/1     Running   0            5m24s
pod/pes2ug23cs906-58b8bf4595-h8qvk  0/1     ContainerCreating 0            8s
pod/pes2ug23cs906-7f6477487-g4cnh  1/1     Running   0            84s
pod/pes2ug23cs906-7f6477487-rhvbq  1/1     Running   0            2m46s
pod/pes2ug23cs906-7f6477487-vhb5x  1/1     Running   0            84s
mona@MonasLaptop:~/PES2UG23CS906$ kubectl rollout history deployment pes2ug23cs906
deployment.apps/pes2ug23cs906
REVISION  CHANGE-CAUSE
1          <none>
2          <none>

mona@MonasLaptop:~/PES2UG23CS906$ kubectl rollout undo deployment/pes2ug23cs906
deployment.apps/pes2ug23cs906 rolled back
mona@MonasLaptop:~/PES2UG23CS906$

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mona@MonasLaptop:~/PES2UG23CS906$ kubectl rollout undo deployment/pes2ug23cs906
deployment.apps/pes2ug23cs906 rolled back
mona@MonasLaptop:~/PES2UG23CS906$ mkdir html
mona@MonasLaptop:~/PES2UG23CS906$ nano html/index.html
mona@MonasLaptop:~/PES2UG23CS906$ nano Dockerfile
mona@MonasLaptop:~/PES2UG23CS906$ eval $(minikube docker-env)

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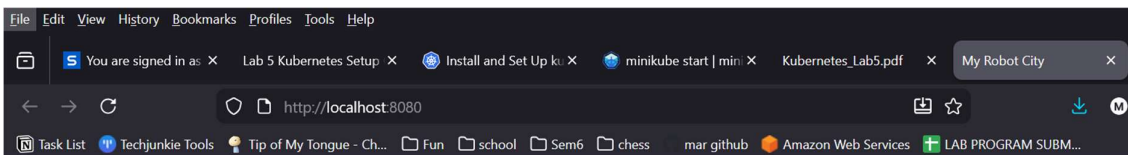
mona@MonasLaptop:~/PES2UG23CS906$ kubectl config set-context --current --namespace=pes2ug23cs906
Context "minikube" modified.
mona@MonasLaptop:~/PES2UG23CS906$ kubectl get deployments
NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-pes2ug23cs906                3/3     3             3            78s
mona@MonasLaptop:~/PES2UG23CS906$ nano pes2ug23cs906-service.yaml
mona@MonasLaptop:~/PES2UG23CS906$ kubectl apply -f pes2ug23cs906-service.yaml
service/pes2ug23cs906-service created
mona@MonasLaptop:~/PES2UG23CS906$ kubectl get services
NAME                                TYPE      CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
pes2ug23cs906-service              ClusterIP  10.107.94.44  <none>        80/TCP    4s
mona@MonasLaptop:~/PES2UG23CS906$ kubectl describe service pes2ug23cs906-service
Name:                             pes2ug23cs906-service
Namespace:                         pes2ug23cs906-beta
Labels:                             <none>
Annotations:                       <none>
Selector:                         app=nginx
Type:                             ClusterIP
IP Family Policy:                 SingleStack
IP Families:                      IPv4
IP:                               10.107.94.44
IPs:                             10.107.94.44
Port:                             <unset> 80/TCP
TargetPort:                       80/TCP
Endpoints:                       10.244.0.16:80,10.244.0.17:80,10.244.0.18:80
Session Affinity:                 None
Internal Traffic Policy:          Cluster
Events:                           <none>
mona@MonasLaptop:~/PES2UG23CS906$ kubectl get endpoints pes2ug23cs906-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME                                ENDPPOINTS   AGE
pes2ug23cs906-service              10.244.0.16:80,10.244.0.17:80,10.244.0.18:80  22s
mona@MonasLaptop:~/PES2UG23CS906$ kubectl delete pod -l app=nginx

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mona@MonasLaptop:~/PES2UG23CS906$ kubectl get endpoints pes2ug23cs906-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME                                ENDPPOINTS   AGE
pes2ug23cs906-service              10.244.0.16:80,10.244.0.17:80,10.244.0.18:80  22s
mona@MonasLaptop:~/PES2UG23CS906$ kubectl delete pod -l app=nginx
pod "nginx-pes2ug23cs906-886b5dff7-5p55q" deleted from pes2ug23cs906-beta namespace
pod "nginx-pes2ug23cs906-886b5dff7-hdxxx" deleted from pes2ug23cs906-beta namespace
pod "nginx-pes2ug23cs906-886b5dff7-wgmqg" deleted from pes2ug23cs906-beta namespace
mona@MonasLaptop:~/PES2UG23CS906$ kubectl get endpoints pes2ug23cs906-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME                                ENDPPOINTS   AGE
pes2ug23cs906-service              10.244.0.19:80,10.244.0.20:80,10.244.0.21:80  41s
mona@MonasLaptop:~/PES2UG23CS906$ kubectl port-forward deployment/nginx-pes2ug23cs906 8080:80
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::1]:8080 -> 80

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Hello from district PES2UG23CS906-beta running in Kubernetes